# Analog Simulation of Burgers Dynamics in a Digital Degenerate Cavity Laser

## 1 Experimental platform

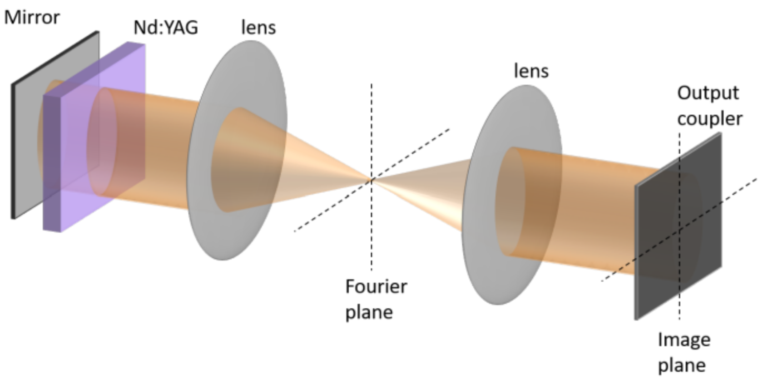
A Digital Degenerate Cavity Laser (DDCL) is a self-imaging $4f$ resonator that supports a large number of transverse laser modes. An aperture mask in a near-field plane defines an array of spatially separated laser sites, while an intracavity spatial light modulator (SLM) in the Fourier plane controls the optical coupling between them.

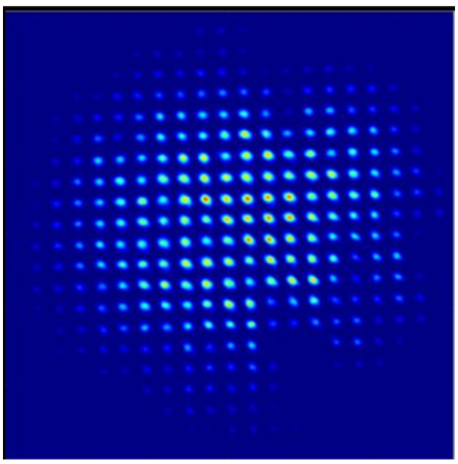
In the present implementation, the nearest-neighbour coupling is programmable in phase and is represented by the complex quantity

$$\kappa = Ke^{i\varphi}, \tag{1}$$

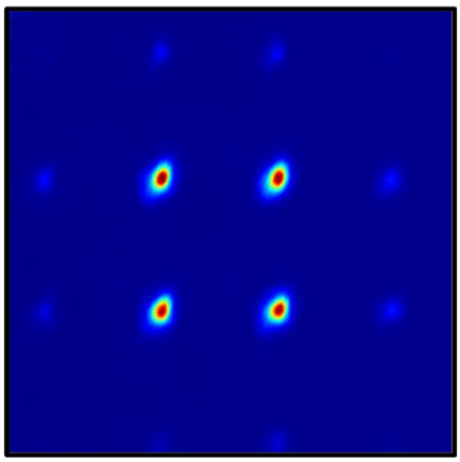
where $K$ is the coupling strength and $\varphi$ is the coupling phase.

As shown in Fig. 1, the DDCL provides both the spatially separated laser array in the near field and access to the collective optical response in the far field. This enables a programmable spatial network of coupled optical oscillators with controllable complex coupling.



(a) Near-field laser array.



(b) Far-field intensity distribution.

Figure 1: Experimental platform. The DDCL provides a self-imaging cavity with programmable intracavity coupling. The near-field plane resolves the individual laser sites, while the far-field distribution reflects the collective coherence of the coupled array.

## 2 From laser rate equations to phase dynamics

The dynamics of the coupled lasers is described by the laser rate equations

$$\dot{E}_m = \frac{1}{\tau_c}\left[(G_m - \alpha_m)E_m + \sum_n \kappa_{mn}E_n\right], \qquad \dot{G}_m = \frac{1}{\tau_f}\left[p_m - G_m - G_m\frac{|E_m|^2}{I_{\text{sat}}}\right]. \tag{2}$$

Writing

$$E_m = A_m e^{-i\theta_m}, \tag{3}$$

the phase dynamics for complex nearest-neighbour coupling contains

$$\dot{\theta}_m = \Omega_m + \frac{K}{\tau_c}\sum_{s=\pm1}\frac{A_{m+s}}{A_m}\sin(\theta_{m+s} - \theta_m - \varphi). \tag{4}$$

The full LRE therefore retain both amplitude and phase dynamics. In the phase-only regime considered here, we assume

$$A_m \simeq A_0, \qquad \Omega_m = \Omega_0. \tag{5}$$

After removing the common global rotation, the phase dynamics reduces to the nearest-neighbour Sakaguchi–Kuramoto model,

$$\boxed{\dot{\theta}_m = K_{\text{eff}}\sum_{s=\pm1}\sin(\theta_{m+s} - \theta_m - \varphi), \qquad K_{\text{eff}} = \frac{K}{\tau_c}.} \tag{6}$$

## 3 Continuum limit: from Kuramoto to Burgers

Introducing $x = ma$ and $\theta_m(t) \to \theta(x,t)$, and assuming small nearest-neighbour phase differences, the continuum expansion of the symmetric Sakaguchi–Kuramoto model gives

$$\theta_t = K_{\text{eff}}a^2\cos\varphi\,\theta_{xx} + K_{\text{eff}}a^2\sin\varphi\,(\theta_x)^2, \tag{7}$$

where the uniform frequency shift has been removed.

Defining the phase-gradient field

$$u = -2K_{\text{eff}}a^2\sin\varphi\,\theta_x, \tag{8}$$

one obtains the one-dimensional viscous Burgers equation

$$\boxed{u_t + u\,u_x = \nu_{\text{eff}}u_{xx}, \qquad \nu_{\text{eff}} = K_{\text{eff}}a^2\cos\varphi.} \tag{9}$$

Thus, the optical phase gradient plays the role of the Burgers field: the nonlinear component of the complex coupling produces advection, while its dissipative component produces effective viscosity.

## 4 Stationary solution

For a stationary state,

$$u_t = 0, \tag{10}$$

and Burgers reduces, after one integration, to

$$\boxed{\frac{du}{dx} + \tan\varphi\,u^2 = C.} \tag{11}$$

The bounded solution has the hyperbolic form

$$u(x) = u_0\tanh\left(\frac{x - x_0}{\ell}\right), \tag{12}$$

where $\ell$ is the characteristic front width and $x_0$ is fixed by the boundary conditions.

For open boundaries, the experimentally reconstructed stationary phase profile therefore provides direct access to the corresponding stationary Burgers solution.

## 5 Experimental correspondence

The main parameters of the optical system map onto the Burgers model as follows:

| Burgers quantity | Experimental realization |
|---|---|
| $x$ | laser-site position |
| $\theta(x,t)$ | optical phase field |
| $u(x,t)$ | phase-gradient field |
| $K_{\text{eff}}$ | coupling strength |
| $\varphi$ | phase of the complex coupling |
| $a$ | spacing between neighbouring laser sites |
| $\nu_{\text{eff}} = K_{\text{eff}} a^2 \cos\varphi$ | effective viscosity |

(13)

The DDCL therefore provides an analog optical realization of Burgers dynamics: the laser array supplies the spatial degrees of freedom, the complex coupling controls the nonlinear and smoothing contributions, and the experimentally reconstructed stationary phase profile provides the stationary Burgers solution.

The mapping is an effective phase-only description of the full LRE and requires approximately equal amplitudes, identical natural frequencies, small nearest-neighbour phase differences, and a long-wavelength continuum limit.